

Avacnes2.5

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```
datos <-read.csv("test.csv")
```

```
##      Id SalePrice economico medio caro
## 1 1461  169277.1          1      0      0
## 2 1462  187758.4          0      1      0
## 3 1463  183583.7          1      0      0
## 4 1464  179317.5          1      0      0
## 5 1465  150730.1          1      0      0
## 6 1466  177151.0          1      0      0

##      Id MSSubClass MSZoning LotFrontage LotArea Street Alley LotShape LandContour
## 1 1 60 RL 65 8450 Pave <NA> Reg Lvl
## 2 2 20 RL 80 9600 Pave <NA> Reg Lvl
## 3 3 60 RL 68 11250 Pave <NA> IR1 Lvl
## 4 4 70 RL 60 9550 Pave <NA> IR1 Lvl
## 5 5 60 RL 84 14260 Pave <NA> IR1 Lvl
## 6 6 50 RL 85 14115 Pave <NA> IR1 Lvl

##      Utilities LotConfig LandSlope Neighborhood Condition1 Condition2 BldgType
## 1 AllPub Inside Gtl CollgCr Norm Norm 1Fam
## 2 AllPub FR2 Gtl Veenker Feedr Norm 1Fam
## 3 AllPub Inside Gtl CollgCr Norm Norm 1Fam
## 4 AllPub Corner Gtl Crawfor Norm Norm 1Fam
## 5 AllPub FR2 Gtl NoRidge Norm Norm 1Fam
## 6 AllPub Inside Gtl Mitchel Norm Norm 1Fam

##      HouseStyle OverallQual OverallCond YearBuilt YearRemodAdd RoofStyle RoofMatl
## 1 2Story 7 5 2003 2003 Gable CompShg
## 2 1Story 6 8 1976 1976 Gable CompShg
## 3 2Story 7 5 2001 2002 Gable CompShg
## 4 2Story 7 5 1915 1970 Gable CompShg
## 5 2Story 8 5 2000 2000 Gable CompShg
## 6 1.5Fin 5 5 1993 1995 Gable CompShg

##      Exterior1st Exterior2nd MasVnrType MasVnrArea ExterQual ExterCond Foundation
## 1 VinylSd VinylSd BrkFace 196 Gd TA PConc
## 2 MetalSd MetalSd None 0 TA TA CBlock
## 3 VinylSd VinylSd BrkFace 162 Gd TA PConc
## 4 Wd Sdng Wd Shng None 0 TA TA BrkTil
## 5 VinylSd VinylSd BrkFace 350 Gd TA PConc
## 6 VinylSd VinylSd None 0 TA TA Wood

##      BsmtQual BsmtCond BsmtExposure BsmtFinType1 BsmtFinSF1 BsmtFinType2
## 1 Gd TA No GLQ 706 Unf
## 2 Gd TA Gd ALQ 978 Unf
## 3 Gd TA Mn GLQ 486 Unf
## 4 TA Gd No ALQ 216 Unf
```

## 5	Gd	TA	Av	GLQ	655	Unf	
## 6	Gd	TA	No	GLQ	732	Unf	
##	BsmtFinSF2	BsmtUnfSF	TotalBsmtSF	Heating	HeatingQC	CentralAir	Electrical
## 1	0	150	856	GasA	Ex	Y	SBrkr
## 2	0	284	1262	GasA	Ex	Y	SBrkr
## 3	0	434	920	GasA	Ex	Y	SBrkr
## 4	0	540	756	GasA	Gd	Y	SBrkr
## 5	0	490	1145	GasA	Ex	Y	SBrkr
## 6	0	64	796	GasA	Ex	Y	SBrkr
##	X1stFlrSF	X2ndFlrSF	LowQualFinSF	GrLivArea	BsmtFullBath	BsmtHalfBath	FullBath
## 1	856	854	0	1710	1	0	2
## 2	1262	0	0	1262	0	1	2
## 3	920	866	0	1786	1	0	2
## 4	961	756	0	1717	1	0	1
## 5	1145	1053	0	2198	1	0	2
## 6	796	566	0	1362	1	0	1
##	HalfBath	BedroomAbvGr	KitchenAbvGr	KitchenQual	TotRmsAbvGrd	Functional	
## 1	1	3	1	Gd	8	Typ	
## 2	0	3	1	TA	6	Typ	
## 3	1	3	1	Gd	6	Typ	
## 4	0	3	1	Gd	7	Typ	
## 5	1	4	1	Gd	9	Typ	
## 6	1	1	1	TA	5	Typ	
##	Fireplaces	FireplaceQu	GarageType	GarageYrBlt	GarageFinish	GarageCars	
## 1	0	<NA>	Attchd	2003	RFn	2	
## 2	1	TA	Attchd	1976	RFn	2	
## 3	1	TA	Attchd	2001	RFn	2	
## 4	1	Gd	Detchd	1998	Unf	3	
## 5	1	TA	Attchd	2000	RFn	3	
## 6	0	<NA>	Attchd	1993	Unf	2	
##	GarageArea	GarageQual	GarageCond	PavedDrive	WoodDeckSF	OpenPorchSF	
## 1	548	TA	TA	Y	0	61	
## 2	460	TA	TA	Y	298	0	
## 3	608	TA	TA	Y	0	42	
## 4	642	TA	TA	Y	0	35	
## 5	836	TA	TA	Y	192	84	
## 6	480	TA	TA	Y	40	30	
##	EnclosedPorch	X3SsnPorch	ScreenPorch	PoolArea	PoolQC	Fence	MiscFeature
## 1	0	0	0	0	<NA>	<NA>	<NA>
## 2	0	0	0	0	<NA>	<NA>	<NA>
## 3	0	0	0	0	<NA>	<NA>	<NA>
## 4	272	0	0	0	<NA>	<NA>	<NA>
## 5	0	0	0	0	<NA>	<NA>	<NA>
## 6	0	320	0	0	<NA>	MnPrv	Shed
##	MiscVal	MoSold	YrSold	SaleType	SaleCondition	SalePrice	economico medio caro
## 1	0	2	2008	WD	Normal	208500	1 0 0
## 2	0	5	2007	WD	Normal	181500	1 0 0
## 3	0	9	2008	WD	Normal	223500	1 0 0
## 4	0	2	2006	WD	Abnorml	140000	1 0 0
## 5	0	12	2008	WD	Normal	250000	1 0 0
## 6	700	10	2009	WD	Normal	143000	1 0 0

Se observa que utilizamos para la variable objetivo tres categorías, económico, caro y medio donde primero calculamos el valor de la casa máxima y el mínimo valor de la casa donde luego lo restamos el valor máximo

con el mínimo y luego dividirlo entre tres para así diferenciar que casa debería de pertenecer en esas tres variables.

Solo se elimina la variable ID, Exterior2ndCBlock, GrLivArea, TotalBsmtSF porque no es importante en estos analisis y evitar el analisis de más variables.

```
summary(modelo_todas_cv)
```

```
##
## Call:
## NULL
##
## Coefficients: (3 not defined because of singularities)
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    2.657e+01  1.654e+07      0      1
## MSSubClass      6.661e-15  1.301e+03      0      1
## MSZoningFV     -1.533e-13  1.889e+05      0      1
## MSZoningRH      4.699e-13  1.872e+05      0      1
## MSZoningRL      3.488e-13  1.608e+05      0      1
## MSZoningRM      4.195e-13  1.505e+05      0      1
## LotFrontage    -1.545e-14  6.903e+02      0      1
## LotArea        -1.213e-16  1.749e+00      0      1
## StreetPave     -3.577e-12  1.904e+05      0      1
## AlleyPave       1.715e-12  7.469e+04      0      1
## LotShapeIR2    -4.465e-13  6.640e+04      0      1
## LotShapeIR3     1.419e-12  1.398e+05      0      1
## LotShapeReg    -6.592e-13  2.516e+04      0      1
## LandContourHLS -2.003e-12  8.051e+04      0      1
## LandContourLow  1.793e-12  1.001e+05      0      1
## LandContourLvl  2.327e-14  5.794e+04      0      1
## UtilitiesNoSeWa 8.389e-12  4.145e+05      0      1
## LotConfigCulDSac -1.878e-12  5.206e+04      0      1
## LotConfigFR2    -2.475e-13  6.332e+04      0      1
## LotConfigFR3    -2.153e-12  1.976e+05      0      1
## LotConfigInside -2.748e-13  2.814e+04      0      1
## LandSlopeMod     5.262e-13  6.254e+04      0      1
## LandSlopeSev     7.730e-12  1.803e+05      0      1
## NeighborhoodBlueste 8.778e-13  3.032e+05      0      1
## NeighborhoodBrDale -2.786e-15  1.725e+05      0      1
## NeighborhoodBrkSide 1.314e-12  1.489e+05      0      1
## NeighborhoodClearCr 2.268e-12  1.446e+05      0      1
## NeighborhoodCollgCr 2.118e-12  1.141e+05      0      1
## NeighborhoodCrawfor 2.427e-12  1.346e+05      0      1
## NeighborhoodEdwards 3.516e-12  1.261e+05      0      1
## NeighborhoodGilbert 1.773e-12  1.207e+05      0      1
## NeighborhoodIDOTRR 5.281e-12  1.688e+05      0      1
## NeighborhoodMeadowV 1.164e-12  1.763e+05      0      1
## NeighborhoodMitchel 1.662e-12  1.286e+05      0      1
## NeighborhoodNames  1.807e-12  1.232e+05      0      1
## NeighborhoodNoRidge 5.046e-12  1.339e+05      0      1
## NeighborhoodNPkVill -5.475e-13  2.213e+05      0      1
## NeighborhoodNridgHt 1.322e-12  1.193e+05      0      1
## NeighborhoodNWames 2.114e-12  1.262e+05      0      1
## NeighborhoodOldTown 5.554e-12  1.510e+05      0      1
## NeighborhoodSawyer 2.579e-12  1.278e+05      0      1
```

## NeighborhoodSawyerW	1.988e-12	1.224e+05	0	1
## NeighborhoodSomerst	1.683e-12	1.418e+05	0	1
## NeighborhoodStoneBr	-2.015e-12	1.319e+05	0	1
## NeighborhoodSWISU	4.673e-12	1.525e+05	0	1
## NeighborhoodTimber	1.144e-12	1.275e+05	0	1
## NeighborhoodVeenker	3.504e-13	1.653e+05	0	1
## Condition1Feedr	4.815e-12	7.859e+04	0	1
## Condition1Norm	4.107e-12	6.596e+04	0	1
## Condition1PosA	5.164e-12	1.571e+05	0	1
## Condition1PosN	4.473e-12	1.169e+05	0	1
## Condition1RR Ae	4.870e-12	1.427e+05	0	1
## Condition1RR An	3.474e-12	1.090e+05	0	1
## Condition1RR Ne	5.202e-12	2.748e+05	0	1
## Condition1RR Nn	-5.933e-13	2.028e+05	0	1
## Condition2Feedr	2.546e-12	3.681e+05	0	1
## Condition2Norm	-4.530e-12	3.183e+05	0	1
## Condition2PosA	7.785e-12	5.831e+05	0	1
## Condition2PosN	6.791e-12	4.484e+05	0	1
## Condition2RR Ae	1.410e-11	8.951e+05	0	1
## Condition2RR An	-4.819e-12	4.955e+05	0	1
## Condition2RR Nn	-3.474e-12	4.250e+05	0	1
## BldgType2fmCon	-2.640e-12	1.965e+05	0	1
## BldgTypeDuplex	-1.713e-12	1.157e+05	0	1
## BldgTypeTwnhs	-4.017e-13	1.580e+05	0	1
## BldgTypeTwnhsE	3.541e-13	1.422e+05	0	1
## HouseStyle1.5Unf	-5.766e-13	1.243e+05	0	1
## HouseStyle1Story	-1.336e-12	6.860e+04	0	1
## HouseStyle2.5Fin	1.080e-11	1.942e+05	0	1
## HouseStyle2.5Unf	-2.510e-12	1.453e+05	0	1
## HouseStyle2Story	-1.723e-12	5.502e+04	0	1
## HouseStyleSFoyer	-8.396e-13	9.788e+04	0	1
## HouseStyleSLvl	-9.990e-13	8.699e+04	0	1
## OverallQual	1.036e-13	1.631e+04	0	1
## OverallCond	5.208e-13	1.399e+04	0	1
## YearBuilt	-4.551e-15	1.208e+03	0	1
## YearRemodAdd	-3.681e-14	8.758e+02	0	1
## RoofStyleGable	4.064e-12	2.901e+05	0	1
## RoofStyleGambrel	4.350e-12	3.173e+05	0	1
## RoofStyleHip	3.940e-12	2.913e+05	0	1
## RoofStyleMansard	3.202e-12	3.369e+05	0	1
## RoofStyleShed	-4.901e-12	5.456e+05	0	1
## RoofMatlCompShg	-3.391e-11	6.426e+05	0	1
## RoofMatlMembran	-3.073e-11	8.608e+05	0	1
## RoofMatlMetal	-3.811e-11	8.403e+05	0	1
## RoofMatlRoll	-2.966e-11	7.506e+05	0	1
## `RoofMatlTar&Grv`	-2.907e-11	7.013e+05	0	1
## RoofMatlWdShake	-3.427e-11	6.880e+05	0	1
## RoofMatlWdShngl	-3.471e-11	6.686e+05	0	1
## Exterior1stAsphShn	-5.949e-12	5.153e+05	0	1
## Exterior1stBrkComm	1.691e-12	4.351e+05	0	1
## Exterior1stBrkFace	1.219e-12	1.994e+05	0	1
## Exterior1stCBlock	8.826e-12	4.274e+05	0	1
## Exterior1stCemntBd	-3.590e-11	3.040e+05	0	1
## Exterior1stHdBoard	-5.367e-13	2.023e+05	0	1

## Exterior1stImStucc	-1.644e-11	4.436e+05	0	1
## Exterior1stMetalSd	4.740e-13	2.288e+05	0	1
## Exterior1stPlywood	-7.452e-13	1.998e+05	0	1
## Exterior1stStone	6.152e-12	3.767e+05	0	1
## Exterior1stStucco	-4.716e-12	2.215e+05	0	1
## Exterior1stVinylSd	2.108e-12	2.081e+05	0	1
## `Exterior1stWd Sdng`	4.409e-12	1.931e+05	0	1
## Exterior1stWdShing	1.537e-12	2.092e+05	0	1
## Exterior2ndAsphShn	4.137e-12	3.485e+05	0	1
## `Exterior2ndBrk Cmn`	6.902e-13	3.151e+05	0	1
## Exterior2ndBrkFace	-1.247e-12	2.060e+05	0	1
## Exterior2ndCBlock	NA	NA	NA	NA
## Exterior2ndCmentBd	3.728e-11	3.001e+05	0	1
## Exterior2ndHdBoard	1.716e-12	1.945e+05	0	1
## Exterior2ndImStucc	2.061e-11	2.252e+05	0	1
## Exterior2ndMetalSd	1.114e-12	2.229e+05	0	1
## Exterior2ndOther	4.558e-12	4.270e+05	0	1
## Exterior2ndPlywood	1.553e-12	1.889e+05	0	1
## Exterior2ndStone	-1.339e-12	2.668e+05	0	1
## Exterior2ndStucco	-5.230e-12	2.143e+05	0	1
## Exterior2ndVinylSd	-6.318e-13	2.010e+05	0	1
## `Exterior2ndWd Sdng`	-2.578e-12	1.870e+05	0	1
## `Exterior2ndWd Shng`	-1.936e-12	1.955e+05	0	1
## MasVnrTypeBrkFace	-2.249e-12	1.076e+05	0	1
## MasVnrTypeNone	-2.203e-12	1.086e+05	0	1
## MasVnrTypeStone	-2.595e-12	1.138e+05	0	1
## MasVnrArea	-3.952e-15	9.123e+01	0	1
## ExterQualFa	8.387e-12	1.741e+05	0	1
## ExterQualGd	7.456e-12	7.594e+04	0	1
## ExterQualTA	9.024e-12	8.380e+04	0	1
## ExterCondFa	5.801e-12	2.824e+05	0	1
## ExterCondGd	1.060e-12	2.699e+05	0	1
## ExterCondPo	9.093e-12	4.948e+05	0	1
## ExterCondTA	2.250e-12	2.693e+05	0	1
## FoundationCBlock	-3.620e-13	4.991e+04	0	1
## FoundationPConc	-2.265e-12	5.381e+04	0	1
## FoundationSlab	8.230e-13	1.233e+05	0	1
## FoundationStone	2.471e-12	1.769e+05	0	1
## FoundationWood	-2.084e-12	2.325e+05	0	1
## BsmtQualFa	-2.740e-12	9.951e+04	0	1
## BsmtQualGd	-3.467e-12	5.389e+04	0	1
## BsmtQualTA	-3.393e-12	6.598e+04	0	1
## BsmtCondGd	-1.943e-12	8.297e+04	0	1
## BsmtCondPo	-1.524e-11	4.703e+05	0	1
## BsmtCondTA	-1.665e-12	6.669e+04	0	1
## BsmtExposureGd	-2.608e-12	4.779e+04	0	1
## BsmtExposureMn	3.545e-14	4.756e+04	0	1
## BsmtExposureNo	-1.427e-13	3.429e+04	0	1
## BsmtFinType1BLQ	5.532e-13	4.386e+04	0	1
## BsmtFinType1GLQ	4.038e-13	3.971e+04	0	1
## BsmtFinType1LwQ	9.449e-13	5.885e+04	0	1
## BsmtFinType1Rec	1.548e-12	4.710e+04	0	1
## BsmtFinType1Unf	4.131e-13	4.550e+04	0	1
## BsmtFinSF1	6.287e-17	7.473e+01	0	1

## BsmtFinType2BLQ	3.962e-13	1.187e+05	0	1
## BsmtFinType2GLQ	-1.447e-12	1.474e+05	0	1
## BsmtFinType2LwQ	-4.191e-12	1.162e+05	0	1
## BsmtFinType2Rec	-4.805e-13	1.118e+05	0	1
## BsmtFinType2Unf	1.102e-13	1.183e+05	0	1
## BsmtFinSF2	2.161e-15	1.343e+02	0	1
## BsmtUnfSF	-1.396e-16	6.293e+01	0	1
## TotalBsmtSF	NA	NA	NA	NA
## HeatingGasA	2.104e-12	3.867e+05	0	1
## HeatingGasW	4.472e-12	4.006e+05	0	1
## HeatingGrav	5.751e-13	4.259e+05	0	1
## HeatingOthW	3.451e-12	4.850e+05	0	1
## HeatingWall	8.756e-12	4.500e+05	0	1
## HeatingQCFA	-1.317e-12	7.416e+04	0	1
## HeatingQCGd	-6.874e-13	3.253e+04	0	1
## HeatingQCPo	-1.435e-12	4.180e+05	0	1
## HeatingQCTA	1.930e-13	3.258e+04	0	1
## CentralAirY	-1.391e-12	6.081e+04	0	1
## ElectricalFuseF	1.177e-13	9.041e+04	0	1
## ElectricalFuseP	-1.714e-12	2.899e+05	0	1
## ElectricalMix	1.564e-12	7.000e+05	0	1
## ElectricalSBrkr	-2.232e-13	4.626e+04	0	1
## X1stFlrSF	-2.889e-16	8.657e+01	0	1
## X2ndFlrSF	-1.529e-15	9.389e+01	0	1
## LowQualFinSF	-4.523e-14	2.954e+02	0	1
## GrLivArea	NA	NA	NA	NA
## BsmtFullBath	-3.463e-13	3.110e+04	0	1
## BsmtHalfBath	-2.140e-12	4.758e+04	0	1
## FullBath	-1.709e-13	3.469e+04	0	1
## HalfBath	9.954e-13	3.297e+04	0	1
## BedroomAbvGr	3.263e-13	2.152e+04	0	1
## KitchenAbvGr	1.719e-12	8.914e+04	0	1
## KitchenQualFa	-2.590e-12	9.784e+04	0	1
## KitchenQualGd	-1.837e-12	5.585e+04	0	1
## KitchenQualTA	-1.554e-12	6.257e+04	0	1
## TotRmsAbvGrd	-3.200e-13	1.504e+04	0	1
## FunctionalMaj2	6.646e-12	2.265e+05	0	1
## FunctionalMin1	-1.694e-12	1.354e+05	0	1
## FunctionalMin2	-2.800e-12	1.357e+05	0	1
## FunctionalMod	-1.147e-11	1.661e+05	0	1
## FunctionalSev	2.264e-14	4.649e+05	0	1
## FunctionalTyp	-2.455e-12	1.176e+05	0	1
## Fireplaces	3.104e-13	2.401e+04	0	1
## FireplaceQuFa	9.691e-12	1.083e+05	0	1
## FireplaceQuGd	9.266e-12	8.392e+04	0	1
## FireplaceQuPo	9.801e-12	1.233e+05	0	1
## FireplaceQuTA	9.822e-12	8.695e+04	0	1
## GarageTypeAttchd	-2.049e-12	1.735e+05	0	1
## GarageTypeBasment	-1.629e-12	2.004e+05	0	1
## GarageTypeBuiltIn	-3.053e-12	1.806e+05	0	1
## GarageTypeCarPort	-3.931e-12	2.305e+05	0	1
## GarageTypeDetchd	-2.283e-12	1.729e+05	0	1
## GarageYrBlt	5.361e-14	8.923e+02	0	1
## GarageFinishRFn	2.004e-13	3.098e+04	0	1

## GarageFinishUnf	3.763e-14	3.773e+04	0	1
## GarageCars	1.250e-12	3.453e+04	0	1
## GarageArea	-6.820e-15	1.224e+02	0	1
## GarageQualFa	-9.740e-12	4.832e+05	0	1
## GarageQualGd	-1.227e-11	4.941e+05	0	1
## GarageQualPo	-1.076e-11	6.125e+05	0	1
## GarageQualTA	-1.024e-11	4.782e+05	0	1
## GarageCondFa	1.361e-11	5.528e+05	0	1
## GarageCondGd	1.559e-11	5.731e+05	0	1
## GarageCondPo	1.534e-11	5.924e+05	0	1
## GarageCondTA	1.328e-11	5.482e+05	0	1
## PavedDriveP	2.144e-12	8.597e+04	0	1
## PavedDriveY	2.028e-13	5.379e+04	0	1
## WoodDeckSF	-1.217e-15	9.257e+01	0	1
## OpenPorchSF	-5.785e-15	1.806e+02	0	1
## EnclosedPorch	-2.526e-14	1.961e+02	0	1
## X3SsnPorch	-2.427e-16	3.520e+02	0	1
## ScreenPorch	-1.085e-14	1.973e+02	0	1
## PoolArea	9.296e-15	4.636e+02	0	1
## PoolQCFa	-2.021e-10	5.004e+05	0	1
## PoolQCGd	-2.018e-10	4.034e+05	0	1
## FenceGdWo	3.809e-12	7.735e+04	0	1
## FenceMnPrv	3.733e-12	5.704e+04	0	1
## FenceMnWw	4.546e-12	1.292e+05	0	1
## MiscFeatureOthr	2.505e-11	1.006e+06	0	1
## MiscFeatureShed	1.930e-11	1.038e+06	0	1
## MiscFeatureTenC	1.812e-11	1.070e+06	0	1
## MiscVal	1.252e-15	6.304e+01	0	1
## MoSold	8.678e-14	3.858e+03	0	1
## YrSold	-1.270e-14	8.101e+03	0	1
## SaleTypeCon	-1.459e-12	2.768e+05	0	1
## SaleTypeConLD	1.039e-12	1.526e+05	0	1
## SaleTypeConLI	-2.007e-12	1.813e+05	0	1
## SaleTypeConLw	-7.088e-13	1.908e+05	0	1
## SaleTypeCWD	-1.570e-12	2.027e+05	0	1
## SaleTypeNew	3.616e-12	2.427e+05	0	1
## SaleTypeOth	3.002e-13	2.255e+05	0	1
## SaleTypeWD	-9.568e-13	6.570e+04	0	1
## SaleConditionAdjLand	-1.266e-12	2.276e+05	0	1
## SaleConditionAlloca	-7.232e-13	1.364e+05	0	1
## SaleConditionFamily	5.142e-13	9.558e+04	0	1
## SaleConditionNormal	2.484e-13	4.570e+04	0	1
## SaleConditionPartial	-4.573e-12	2.333e+05	0	1
## SalePrice	5.242e-17	5.647e-01	0	1
## economico	-5.313e+01	2.113e+05	0	1
## medio	-5.313e+01	1.853e+05	0	1
##				
## (Dispersion parameter for binomial family taken to be 1)				
##				
## Null deviance: 9.9264e+01 on 1459 degrees of freedom				
## Residual deviance: 8.4703e-09 on 1214 degrees of freedom				
## AIC: 492				
##				
## Number of Fisher Scoring iterations: 25				

El modelo de regresión logística con validación cruzada mostró que ninguna de las variables es significativa, ya que los valores de z son 0 y los valores de $\Pr(>|z|)$ son 1, lo que indica que las variables no están contribuyendo de manera relevante. Esto podría ser causado por la multicolinealidad entre las variables, debido a que muchas de ellas están altamente correlacionadas, lo que afecta la estimación de los coeficientes. Otra posible causa es el exceso de variables en relación con el número de observaciones, lo que podría llevar a un sobreajuste lo cual es más probable con los modelos realizadas con anterioridad.

Los valores de p de 1 son un gran indicativo de la falta de significancia en las variables sobre el modelo. Esto es debido a la gran multicolinealidad que se presenta en el conjunto de datos, además de que ninguna variable resulta significativa para la predicción de las casas “caras”.

```
## Call:
## nnet::multinom(formula = .outcome ~ ., data = dat, decay = param$decay,
##   trace = FALSE)
##
## Coefficients:
##      (Intercept)  MSSubClass MSZoningFV MSZoningRH  MSZoningRL MSZoningRM
## economico -0.01714893  0.001101410  0.1530568  0.2957217 -0.05689274 -0.2604506
## medio      0.01312096 -0.006694639  0.3162988 -0.4647028  0.82131934  1.1182811
##      LotFrontage    LotArea StreetPave  AlleyPave LotShapeIR2
## economico -0.01299268  0.0001491241 -0.5441204  0.20742026 -0.2246062
## medio      -0.01687481  0.0001036350  1.9050850 -0.02189445 -0.5503255
##      LotShapeIR3 LotShapeReg LandContourHLS LandContourLow LandContourLvl
## economico  1.1490900 -0.03284831   -0.28870227    0.5214846    0.3077974
## medio       0.9912686 -0.39390725   -0.06627513    0.6278728    0.1867241
##      UtilitiesNoSeWa LotConfigCulDSac LotConfigFR2 LotConfigFR3
## economico   -0.3790245   -0.4303433  -0.29089932  -0.4764919
## medio        0.7074540   -0.2424822  -0.09335567   0.1178051
##      LotConfigInside LandSlopeMod LandSlopeSev NeighborhoodBlueste
## economico   -0.2662634  -0.1026328   0.4810144    0.1771096
## medio       -0.3429775   0.1027424  -0.2696511   -0.3309488
##      NeighborhoodBrDale NeighborhoodBrkSide NeighborhoodClearCr
## economico    0.09232272    0.4122681    0.8579764
## medio       -0.05203416    0.2004710    0.3427614
##      NeighborhoodCollgCr NeighborhoodCrawfor NeighborhoodEdwards
## economico   -0.6141003    0.8707533    0.07641929
## medio        0.3972050    0.1707178   -0.09984035
##      NeighborhoodGilbert NeighborhoodIDOTRR NeighborhoodMeadowV
## economico   -0.20657514   -0.2520061   -0.34360850
## medio       -0.04037338   -0.4225894    0.07932858
##      NeighborhoodMitchel NeighborhoodNames NeighborhoodNoRidge
## economico   -0.01416389    0.0207002   -0.3561949
## medio       -0.10715137   -0.1621490    0.3524228
##      NeighborhoodNPkVill NeighborhoodNridgHt NeighborhoodNWames
## economico   -0.03353481   -0.090126027  -0.1220062
## medio        0.56598072    0.006903909    0.1874474
##      NeighborhoodOldTown NeighborhoodSawyer NeighborhoodSawyerW
## economico    0.07742748    0.1042675   -0.3875433
## medio       -0.27604055    0.7234797   -0.3815222
##      NeighborhoodSomerst NeighborhoodStoneBr NeighborhoodSWISU
## economico    0.1984696   -0.88008644    0.30924458
## medio       -0.2794237    0.03543834   -0.07452418
##      NeighborhoodTimber NeighborhoodVeenker Condition1Feedr Condition1Norm
## economico    0.4773896    0.54821107    0.3191159    0.05091740
## medio        0.4860941    0.05034178   -0.1661514    0.03322942
```


##	Condition1PosA	Condition1PosN	Condition1RR Ae	Condition1RR An
## economico	0.4552429	0.2413831	0.4151138	-0.07765070
## medio	0.1544174	0.3099631	-0.4416228	-0.05919175
##	Condition1RR Ne	Condition1RR Nn	Condition2Feedr	Condition2Norm
## economico	-0.52754577	-0.3865589	0.601042	0.4077817
## medio	0.01462837	-0.4298720	-1.350585	0.8336800
##	Condition2PosA	Condition2PosN	Condition2RR Ae	Condition2RR An
## economico	-0.4121201	-0.53711508	-0.1980976	-0.20494656
## medio	0.1712187	0.07148083	0.4430674	0.08656668
##	Condition2RR Nn	BldgType2fmCon	BldgTypeDuplex	BldgTypeTwnhs
## economico	0.10872284	-0.2213977	-0.01106503	-0.3203615
## medio	-0.07571559	0.2348846	-0.27582705	0.6400671
##	BldgTypeTwnhsE	HouseStyle1.5Unf	HouseStyle1Story	HouseStyle2.5Fin
## economico	-0.12936422	0.3139577	-0.03307198	0.2366385
## medio	-0.06738761	-0.5213995	-0.38632283	0.4281893
##	HouseStyle2.5Unf	HouseStyle2Story	HouseStyleSFoyer	HouseStyleSLvl
## economico	-0.3161256	-0.008236034	0.14599529	-0.3603264
## medio	-0.3267774	0.356919350	-0.07447617	-0.2552559
##	OverallQual	OverallCond	YearBuilt	YearRemodAdd
## economico	0.4098179	0.3226525	0.01518136	0.005503871
## medio	0.1220051	0.7728217	0.03064422	-0.016057561
##	RoofStyleGambrel	RoofStyleHip	RoofStyleMansard	RoofStyleShed
## economico	0.053505473	-0.4552475	-0.001746087	-0.2011856
## medio	0.002144258	-1.0616212	0.600209922	0.4568513
##	RoofMatlCompShg	RoofMatlMembran	RoofMatlMetal	RoofMatlRoll
## economico	0.03653475	0.32026835	0.2974809	0.2993835
## medio	0.43899540	-0.01349971	0.1097386	0.1285226
##	RoofMatlTar&Grv`	RoofMatlWdShake	RoofMatlWdShngl	Exterior1stAsphShn
## economico	-0.2127485	-0.19032503	-0.4767200	-0.3179992
## medio	0.3525885	-0.07901624	-0.4340175	0.1326490
##	Exterior1stBrkComm	Exterior1stBrkFace	Exterior1stCBlock	
## economico	-0.16359160	0.48116447	0.06121009	
## medio	0.07876888	-0.01821254	-0.09771173	
##	Exterior1stCemntBd	Exterior1stHdBoard	Exterior1stImStucc	
## economico	-1.5019220	-0.1172292	0.1150858	
## medio	-0.7030497	-0.1473618	-0.1137626	
##	Exterior1stMetalSd	Exterior1stPlywood	Exterior1stStone	
## economico	0.4450422	0.7053514	0.1203959	
## medio	-0.1514494	-0.1794610	-0.3794658	
##	Exterior1stStucco	Exterior1stVinylSd	Exterior1stWd Sdng`	
## economico	-0.5637144	0.7217669	0.2131426	
## medio	0.7357022	-0.2902308	-0.1138478	
##	Exterior1stWdShing	Exterior2ndAsphShn	Exterior2ndBrk Cmn`	
## economico	0.52076672	0.3276143	0.06730893	
## medio	0.01953439	-0.2802776	0.31734865	
##	Exterior2ndBrkFace	Exterior2ndCBlock	Exterior2ndCmentBd	
## economico	-0.2195023	0.06121009	0.84910659	
## medio	0.1078082	-0.09771173	0.02124405	
##	Exterior2ndHdBoard	Exterior2ndImStucc	Exterior2ndMetalSd	
## economico	-0.7065934	-0.07080637	-0.2893852	
## medio	-0.1985426	-0.09238393	-0.1014440	
##	Exterior2ndOther	Exterior2ndPlywood	Exterior2ndStone	
## economico	0.0537975	-0.1877785	-0.3195776	
## medio	-0.1672481	-0.2087765	0.3462856	

##	Exterior2ndStucco	Exterior2ndVinylSd	`Exterior2ndWd Sdng`		
## economico	0.7306478	-0.11743038	0.004325423		
## medio	-0.1457104	-0.01608508	0.038003628		
##	`Exterior2ndWd Shng` MasVnrTypeBrkFace MasVnrTypeNone MasVnrTypeStone				
## economico	-0.6456399	-0.01680262	0.33396639	-0.06364656	
## medio	-0.1597615	-0.28403460	0.09018723	0.16693987	
##	MasVnrArea	ExterQualFa	ExterQualGd	ExterQualTA	ExterCondFa
## economico	0.001583670	-0.07678328	0.4148993	-0.3057784	0.2202689
## medio	0.001607949	-0.87318144	-0.9032618	-1.1099690	-0.1694694
##	ExterCondGd	ExterCondPo	ExterCondTA	FoundationCBlock	FoundationPConc
## economico	0.36855995	0.04833628	0.1493948	0.1714171	0.4462299
## medio	-0.03047594	0.14765172	-0.1995359	0.2197555	0.7311633
##	FoundationSlab	FoundationStone	FoundationWood	BsmtQualFa	BsmtQualGd
## economico	-0.6480286	1.438981	0.6807202	-0.4766127	0.1333541
## medio	-0.1745766	0.651968	-0.6031166	0.1479070	0.6127214
##	BsmtQualTA	BsmtCondGd	BsmtCondPo	BsmtCondTA	BsmtExposureGd
## economico	0.1831356	0.4493871	0.3855699	-0.3305763	-0.4371681
## medio	-0.2544266	-0.4158386	-0.7869508	0.4019760	-0.4823639
##	BsmtExposureMn	BsmtExposureNo	BsmtFinType1BLQ	BsmtFinType1GLQ	
## economico	0.3316429	0.2872012	0.03156766	0.1725523	
## medio	-0.2854998	-0.3415710	0.12686212	0.4102017	
##	BsmtFinType1LwQ	BsmtFinType1Rec	BsmtFinType1Unf	BsmtFinSF1	
## economico	0.08289960	-0.1537751	-0.1133463	-0.0003044563	
## medio	-0.06443489	-0.2297761	0.1163508	-0.0004281523	
##	BsmtFinType2BLQ	BsmtFinType2GLQ	BsmtFinType2LwQ	BsmtFinType2Rec	
## economico	0.2830420	0.01488595	0.4146735	0.1552275	
## medio	-0.1152676	-0.04784530	0.2965416	0.5653399	
##	BsmtFinType2Unf	BsmtFinSF2	BsmtUnfSF	TotalBsmtSF	HeatingGasA
## economico	0.2371898	-0.0001721706	-0.0003553337	-0.0008319681	-0.7506293
## medio	0.5097313	0.0006826186	-0.0004627562	-0.0002082921	-0.3838489
##	HeatingGasW	HeatingGrav	HeatingOthW	HeatingWall	HeatingQCFa
## economico	0.5109077	-0.2522827	0.06104162	0.11413934	0.06258231
## medio	0.8908967	-0.2538473	-0.06823086	-0.09284914	-0.05108432
##	HeatingQCGd	HeatingQCPo	HeatingQCTA	CentralAirY	ElectricalFuseF
## economico	0.1055135	-0.1413208	-0.12858786	0.1073664	-0.04027725
## medio	-0.1480589	0.1183587	-0.08465793	0.4735902	0.28771567
##	ElectricalFuseP	ElectricalMix	ElectricalSBrkr	X1stFlrSF	
## economico	0.09306355	0.2829388	0.4308974	9.467992e-05	
## medio	-0.08511168	-0.3774419	0.6376577	-2.985019e-04	
##	X2ndFlrSF	LowQualFinSF	GrLivArea	BsmtFullBath	BsmtHalfBath
## economico	-0.0009751435	0.003073956	0.002193534	0.5262366	0.04053264
## medio	-0.0018323491	0.003778733	0.001647851	-0.2288404	-0.05947129
##	FullBath	HalfBath	BedroomAbvGr	KitchenAbvGr	KitchenQualFa
## economico	0.1930876	0.03125897	0.8890714	0.1752478	-0.130697
## medio	0.4375040	0.02837515	-0.8042966	0.2776410	-0.288850
##	KitchenQualGd	KitchenQualTA	TotRmsAbvGrd	FunctionalMaj2	
## economico	0.23097437	0.2996015	-0.6577827	1.8515709	
## medio	0.08221661	-0.3058388	0.3079607	-0.9223682	
##	FunctionalMin1	FunctionalMin2	FunctionalMod	FunctionalSev	
## economico	-0.2850374	0.01360182	-1.3558075	0.3758713	
## medio	0.3242594	0.58516888	-0.2171951	-0.7033537	
##	FunctionalTyp	Fireplaces	FireplaceQuFa	FireplaceQuGd	FireplaceQuPo
## economico	-0.2049661	0.4809739	-0.4609171	-1.0415242	-0.7615189
## medio	0.6885652	0.5366603	-0.3631750	-0.4783431	-0.2812913

```

##      FireplaceQuTA GarageTypeAttchd GarageTypeBasment GarageTypeBuiltIn
## economico      -0.8728485      0.07246153      0.1371366      0.04449576
## medio          -0.7370135      -0.39216198      0.7122471      0.14670151
##      GarageTypeCarPort GarageTypeDetchd GarageYrBlt GarageFinishRfN
## economico      -0.1610016      0.006406593 0.0002712129      0.02656806
## medio          -0.7544620      -0.184441077 -0.0019906569      -0.17405334
##      GarageFinishUnf GarageCars GarageArea GarageQualFa GarageQualGd
## economico      0.33036404 0.18366247 -0.0003501396 -0.36424658 -0.2193659
## medio          0.02684333 -0.02114018 0.0001952360 0.06002847 0.2364312
##      GarageQualPo GarageQualTA GarageCondFa GarageCondGd GarageCondPo
## economico      1.0945677 -0.67749935 -0.3563443 -0.4375111 0.7355742
## medio          -0.8563868 -0.03859975 0.5380417 0.1596827 -0.7560980
##      GarageCondTA PavedDriveP PavedDriveY WoodDeckSF OpenPorchSF
## economico      -0.2895975 -0.7865759 -0.3764457 0.0003986864 -0.001866176
## medio          0.3487040 -0.5764269 -0.4030238 -0.0013835478 -0.002585929
##      EnclosedPorch X3SsnPorch ScreenPorch PoolArea PoolQCFA
## economico      0.003546126 0.012549937 -0.001965906 0.003059456 -0.02054309
## medio          0.007759641 0.007561441 0.001412170 0.003931773 0.37359978
##      PoolQCGd FenceGdWo FenceMnPrv FenceMnWw MiscFeature0thr
## economico      -0.8259283 -0.3665405 -0.34011755 -0.05119467 0.02097904
## medio          -0.5891223 -0.2126394 0.05193031 -0.20175737 0.07356371
##      MiscFeatureShed MiscFeatureTenC MiscVal MoSold
## economico      -0.1090484 -0.04283081 0.0005769711 0.0008385918
## medio          -0.7836459 0.84486441 -0.0005527146 -0.0244102735
##      YrSold SaleTypeCon SaleTypeConLD SaleTypeConLI SaleTypeConLw
## economico      -0.01980608 0.4177099 0.5036697 0.2127623 -0.02612279
## medio          -0.01437454 -0.3320231 0.1212872 0.9382414 0.24388227
##      SaleTypeCWD SaleTypeNew SaleType0th SaleTypeWD SaleConditionAdjLand
## economico      0.4006678 0.2053054 0.3017478 -0.09414143 -0.5831784
## medio          0.3024123 0.1679612 -0.1213645 0.31955287 0.6171371
##      SaleConditionAlloca SaleConditionFamily SaleConditionNormal
## economico      0.5008348 0.02074259 -0.03052446
## medio          -1.3604822 0.32027298 0.34547539
##      SaleConditionPartial SalePrice economico medio caro
## economico      -0.1129704 -2.554907e-05 12.480314 -4.934031 -7.563431
## medio          0.3138578 -2.006937e-05 -1.498526 12.915822 -11.404175
##
## Std. Errors:
##      (Intercept) MSSubClass MSZoningFV MSZoningRH MSZoningRL
## economico      8.006962e-06 0.1557881 8.744219e-05 2.949224e-05 0.0005235057
## medio          7.604434e-06 0.0984562 1.521733e-04 2.856644e-06 0.0004211864
##      MSZoningRM LotFrontage LotArea StreetPave AlleyPave
## economico      0.0004922520 0.03414598 0.003347460 4.542891e-05 6.294235e-05
## medio          0.0003514056 0.02703459 0.003343605 7.592089e-06 1.829386e-05
##      LotShapeIR2 LotShapeIR3 LotShapeReg LandContourHLS LandContourLow
## economico      1.056805e-04 1.707236e-05 0.0002734384 0.0001101008 4.635251e-05
## medio          9.914008e-05 1.661205e-05 0.0003658016 0.0001405420 2.793003e-05
##      LandContourLvl UtilitiesNoSeWa LotConfigCulDSac LotConfigFR2
## economico      0.0002503898 0.0002355811 0.0003112939 9.306908e-05
## medio          0.0001641265 0.0002624652 0.0003122132 1.173668e-04
##      LotConfigFR3 LotConfigInside LandSlopeMod LandSlopeSev
## economico      1.239420e-05 0.0002019750 0.0001255374 2.89539e-05
## medio          1.148346e-05 0.0002224911 0.0001575924 3.88484e-05
##      NeighborhoodBlueste NeighborhoodBrDale NeighborhoodBrkSide

```

## economico	2.455551e-06	1.246459e-04	1.892147e-04		
## medio	1.634291e-06	7.567216e-05	8.284681e-05		
##	NeighborhoodClearCr	NeighborhoodCollgCr	NeighborhoodCrawfor		
## economico	8.718677e-05	0.0003883228	7.221292e-05		
## medio	7.269436e-05	0.0002967880	1.404745e-04		
##	NeighborhoodEdwards	NeighborhoodGilbert	NeighborhoodIDOTRR		
## economico	1.020884e-04	0.0001357913	1.919710e-04		
## medio	6.453488e-05	0.0001131858	7.639078e-05		
##	NeighborhoodMeadowV	NeighborhoodMitchel	NeighborhoodNames		
## economico	0.0003197793	5.568615e-05	0.0002894844		
## medio	0.0002294299	3.531849e-05	0.0001526972		
##	NeighborhoodNoRidge	NeighborhoodNPkVill	NeighborhoodNridgHt		
## economico	8.691373e-05	4.645447e-05	0.0002374752		
## medio	6.047711e-05	2.965833e-05	0.0002489233		
##	NeighborhoodNWAmes	NeighborhoodOldTown	NeighborhoodSawyer		
## economico	0.0001929931	0.0002923957	1.181406e-04		
## medio	0.0001272635	0.0001917203	6.481616e-05		
##	NeighborhoodSawyerW	NeighborhoodSomerst	NeighborhoodStoneBr		
## economico	5.080920e-05	0.0001033116	0.00017405		
## medio	4.387882e-05	0.0001726743	0.00018882		
##	NeighborhoodSWISU	NeighborhoodTimber	NeighborhoodVeenker		
## economico	2.150230e-05	0.0002282526	1.180398e-05		
## medio	1.921472e-05	0.0002365110	1.397384e-05		
##	Condition1Feedr	Condition1Norm	Condition1PosA	Condition1PosN	
## economico	7.873359e-05	0.0001244117	4.426290e-05	2.758248e-05	
## medio	5.622046e-05	0.0001015582	7.150297e-05	4.187672e-05	
##	Condition1RR Ae	Condition1RR An	Condition1RR Ne	Condition1RR Nn	
## economico	8.794045e-06	9.193757e-05	1.713039e-05	3.932509e-05	
## medio	1.515519e-05	5.546320e-05	1.591023e-05	7.726112e-06	
##	Condition2Feedr	Condition2Norm	Condition2PosA	Condition2PosN	
## economico	9.033212e-06	6.918377e-05	8.870379e-09	2.435800e-05	
## medio	2.530654e-06	6.773125e-05	4.760552e-05	2.200284e-05	
##	Condition2RR Ae	Condition2RR An	Condition2RR Nn	BldgType2fmCon	
## economico	2.192364e-07	1.339291e-06	1.009610e-05	2.013319e-04	
## medio	3.441648e-07	8.055687e-07	6.019827e-06	3.256161e-05	
##	BldgTypeDuplex	BldgTypeTwnhs	BldgTypeTwnhsE	HouseStyle1.5Unf	
## economico	1.284857e-04	0.0004748924	0.0006239287	2.178026e-05	
## medio	5.679441e-05	0.0003829690	0.0004158675	1.676275e-05	
##	HouseStyle1Story	HouseStyle2.5Fin	HouseStyle2.5Unf	HouseStyle2Story	
## economico	0.0004815599	4.321862e-05	1.426519e-04	0.0002153103	
## medio	0.0003152952	4.450851e-05	9.610759e-05	0.0001335781	
##	HouseStyleSFoyer	HouseStyleSLvl	OverallQual	OverallCond	YearBuilt
## economico	0.0002532111	0.0002550265	0.0007876906	0.0009062944	0.01682424
## medio	0.0002008359	0.0002250583	0.0007470030	0.0006247071	0.01280595
##	YearRemodAdd	RoofStyleGable	RoofStyleGambrel	RoofStyleHip	
## economico	0.013062159	0.0001202384	5.627251e-05	0.0001271590	
## medio	0.008334655	0.0002312783	6.563805e-05	0.0001594939	
##	RoofStyleMansard	RoofStyleShed	RoofMatlCompShg	RoofMatlMembran	
## economico	3.920651e-05	1.501972e-06	5.532289e-05	2.400228e-07	
## medio	5.602619e-05	1.622848e-06	8.081182e-05	2.845022e-07	
##	RoofMatlMetal	RoofMatlRoll	RoofMatlTar&Grv`	RoofMatlWdShake	
## economico	1.259127e-07	7.915787e-06	1.377532e-05	4.081561e-05	
## medio	1.511586e-07	2.430316e-06	1.142554e-05	5.998133e-05	
##	RoofMatlWdShngl	Exterior1stAsphShn	Exterior1stBrkComm		

## economico	4.787567e-06	4.557883e-07	3.993340e-06
## medio	2.371811e-05	1.727826e-07	1.871246e-06
##	Exterior1stBrkFace	Exterior1stCBlock	Exterior1stCemntBd
## economico	6.409669e-05	2.551975e-05	0.0003734905
## medio	8.981073e-05	2.820260e-05	0.0002954096
##	Exterior1stHdBoard	Exterior1stImStucc	Exterior1stMetalSd
## economico	0.0002849773	6.037610e-06	0.0003132943
## medio	0.0003455778	6.222043e-06	0.0002011090
##	Exterior1stPlywood	Exterior1stStone	Exterior1stStucco
## economico	7.472799e-05	4.665611e-05	6.002219e-05
## medio	6.455129e-05	5.022823e-05	7.041421e-05
##	Exterior1stVinylSd	Exterior1stWd Sdng	Exterior1stWdShing
## economico	0.0002587044	0.0002775036	7.058473e-05
## medio	0.0003130797	0.0001228940	7.230134e-05
##	Exterior2ndAsphShn	Exterior2ndBrk Cmn	Exterior2ndBrkFace
## economico	1.387971e-06	2.479224e-05	3.718113e-05
## medio	1.346984e-06	2.624154e-05	2.808158e-05
##	Exterior2ndCBlock	Exterior2ndCmentBd	Exterior2ndHdBoard
## economico	2.551975e-05	0.0003730714	0.0002769573
## medio	2.820260e-05	0.0002850796	0.0003147095
##	Exterior2ndImStucc	Exterior2ndMetalSd	Exterior2ndOther
## economico	1.637478e-05	0.0003236526	2.865251e-09
## medio	1.604781e-05	0.0002029960	6.542195e-06
##	Exterior2ndPlywood	Exterior2ndStone	Exterior2ndStucco
## economico	0.0001980831	4.858887e-05	5.343224e-05
## medio	0.0000877670	6.306255e-05	6.032283e-05
##	Exterior2ndVinylSd	Exterior2ndWd Sdng	Exterior2ndWd Shng
## economico	0.0002541827	2.331461e-04	0.0001438151
## medio	0.0003179308	8.282782e-05	0.0001137529
##	MasVnrTypeBrkFace	MasVnrTypeNone	MasVnrTypeStone
## economico	0.0003416164	0.0003874761	8.237495e-05
## medio	0.0002361880	0.0003194825	1.433766e-04
##	ExterQualFa	ExterQualGd	ExterQualTA
## economico	9.284458e-05	0.0002583483	0.0003143904
## medio	3.317203e-05	0.0003969662	0.0002707649
##	ExterCondPo	ExterCondTA	FoundationCBlock
## economico	4.066483e-05	1.319110e-04	0.0002392530
## medio	4.559186e-07	7.243108e-05	0.0002452191
##	FoundationSlab	FoundationStone	FoundationWood
## economico	5.840562e-05	2.706162e-05	3.149216e-06
## medio	3.728269e-05	4.025096e-06	9.895527e-07
##	BsmtQualFa	BsmtQualGd	BsmtQualTA
## economico	0.0004134461	0.0004058647	2.711733e-05
## medio	0.0004588052	0.0003120336	1.318607e-04
##	BsmtExposureGd	BsmtExposureMn	BsmtExposureNo
## economico	0.0002380998	0.0001190622	0.0004469458
## medio	0.0001378178	0.0001839419	0.0002793317
##	BsmtFinType1GLQ	BsmtFinType1LwQ	BsmtFinType1Rec
## economico	0.0003384430	0.0001584538	0.0002036833
## medio	0.0003383033	0.0001117593	0.0001416001
##	BsmtFinType1Unf	BsmtFinType2BLQ	BsmtFinType2GLQ
## economico	0.03417969	1.140216e-04	5.357128e-05
## medio	0.04368389	7.548475e-05	7.575708e-05
##	BsmtFinType2Rec	BsmtFinType2Unf	BsmtFinSF2
##	BsmtUnfSF	TotalBsmtSF	

## economico	0.0002638790	0.0002267775	0.05515344	0.02958684	0.05229199
## medio	0.0003629019	0.0002364323	0.07152798	0.03903486	0.07320409
##	HeatingGasA	HeatingGasW	HeatingGrav	HeatingOthW	HeatingWall
## economico	4.977862e-05	4.181255e-05	1.297872e-05	6.809691e-06	2.362325e-05
## medio	3.045633e-05	1.709898e-05	1.839082e-06	7.956691e-06	1.024473e-05
##	HeatingQCFa	HeatingQCGd	HeatingQCPo	HeatingQCTA	CentralAirY
## economico	9.998353e-05	0.0002683178	1.132878e-05	0.0003166748	0.0002945485
## medio	3.521856e-05	0.0001672985	8.816188e-06	0.0003830492	0.0000846418
##	ElectricalFuseF	ElectricalFuseP	ElectricalMix	ElectricalSBkr	
## economico	1.055291e-04	9.675324e-06	4.302592e-08	0.0003616570	
## medio	5.824351e-05	9.796053e-06	1.933887e-11	0.0001823056	
##	X1stFlrSF	X2ndFlrSF	LowQualFinSF	GrLivArea	BsmtFullBath
## economico	0.05732055	0.04601704	0.09629693	0.04403392	0.0002780259
## medio	0.07715615	0.04983829	0.08793452	0.04828416	0.0002176794
##	BsmtHalfBath	FullBath	HalfBath	BedroomAbvGr	KitchenAbvGr
## economico	0.0001917648	0.0002489269	0.0002537577	0.0008718895	2.142726e-04
## medio	0.0001916004	0.0001308022	0.0002864086	0.0007487350	9.736856e-05
##	KitchenQualFa	KitchenQualGd	KitchenQualTA	TotRmsAbvGrd	FunctionalMaj2
## economico	1.011879e-04	0.0003661255	0.0004653296	0.0011797964	2.837907e-07
## medio	5.142075e-05	0.0003053397	0.0003526669	0.0008085687	4.273614e-08
##	FunctionalMin1	FunctionalMin2	FunctionalMod	FunctionalSev	
## economico	5.742257e-05	5.098698e-05	6.623612e-05	1.488455e-07	
## medio	5.146807e-05	2.299805e-05	1.076459e-04	2.491323e-07	
##	FunctionalTyp	Fireplaces	FireplaceQuFa	FireplaceQuGd	FireplaceQuPo
## economico	0.0001234834	0.0005149597	1.950680e-05	0.0002912886	3.682474e-05
## medio	0.0001404665	0.0002073631	1.175888e-05	0.0002146868	1.526448e-05
##	FireplaceQuTA	GarageTypeAttchd	GarageTypeBasment	GarageTypeBuiltIn	
## economico	0.0002793807	0.0003430456	0.0001780008	1.08285e-04	
## medio	0.0001758973	0.0002762439	0.0002294329	9.96254e-05	
##	GarageTypeCarPort	GarageTypeDetchd	GarageYrBlt	GarageFinishRfn	
## economico	9.24599e-05	0.000325671	0.01139246	0.0002645142	
## medio	1.58306e-05	0.000317826	0.01137765	0.0002272131	
##	GarageFinishUnf	GarageCars	GarageArea	GarageQualFa	GarageQualGd
## economico	0.0001957925	0.0002702174	0.07561051	1.004408e-04	5.317039e-05
## medio	0.0002902356	0.0003751510	0.11276245	6.228461e-05	5.049581e-05
##	GarageQualPo	GarageQualTA	GarageCondFa	GarageCondGd	GarageCondPo
## economico	7.286251e-07	1.435991e-04	1.071077e-04	5.507786e-05	6.378271e-05
## medio	9.161089e-08	5.131837e-05	4.905453e-05	1.344296e-05	5.893577e-06
##	GarageCondTA	PavedDriveP	PavedDriveY	WoodDeckSF	OpenPorchSF
## economico	1.797251e-04	4.588999e-05	1.447329e-04	0.1218680	0.1917451
## medio	4.789179e-05	3.132719e-05	6.906853e-05	0.1696755	0.2796749
##	EnclosedPorch	X3SsnPorch	ScreenPorch	PoolArea	PoolQCFa
## economico	0.1459642	0.002136383	0.2328127	0.03136260	5.349700e-05
## medio	0.1181985	0.002333017	0.2788786	0.03109531	5.470326e-05
##	PoolQCGd	FenceGdWo	FenceMnPrv	FenceMnWw	MiscFeature0thr
## economico	5.431176e-05	5.774394e-05	1.116181e-04	2.051841e-05	3.792663e-05
## medio	5.604732e-05	2.522631e-05	3.853601e-05	8.890498e-06	4.841111e-06
##	MiscFeatureShed	MiscFeatureTenC	MiscVal	MoSold	YrSold
## economico	4.054199e-05	5.310364e-05	0.09422977	0.0017109937	0.01603303
## medio	5.562294e-05	5.439328e-05	0.13373654	0.0009669727	0.01532265
##	SaleTypeCon	SaleTypeConLD	SaleTypeConLI	SaleTypeConLw	SaleTypeCWD
## economico	8.516569e-07	5.422627e-05	4.483628e-06	1.280649e-05	4.034112e-05
## medio	8.020523e-07	4.351956e-06	7.011543e-06	1.269163e-05	1.370041e-05
##	SaleTypeNew	SaleTypeOth	SaleTypeWD	SaleConditionAdjLand	

```

## economico 8.129496e-05 3.272053e-06 0.0003539961      2.217397e-05
## medio     1.832100e-04 6.004988e-07 0.0002896359      1.051574e-05
##           SaleConditionAlloca SaleConditionFamily SaleConditionNormal
## economico      2.561715e-05      4.786098e-05      0.0002803497
## medio          1.884204e-05      1.498812e-05      0.0002716672
##           SaleConditionPartial SalePrice      economico      medio
## economico      8.667338e-05 0.0003557239 3.817706e-05 0.0000414717
## medio          1.831999e-04 0.0004087076 1.180959e-04 0.0002148406
##           caro
## economico 4.667592e-05
## medio     1.167636e-04
##
## Residual Deviance: 0.1498268
## AIC: 984.1498

## Confusion Matrix and Statistics
##
##           Reference
## Prediction  caro economico medio
## caro        8           0         0
## economico   2          1010       0
## medio        0           0        439
##
## Overall Statistics
##
##           Accuracy : 0.9986
##           95% CI : (0.9951, 0.9998)
##           No Information Rate : 0.6923
##           P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.9968
##
## Mcnemar's Test P-Value : NA
##
## Statistics by Class:
##
##           Class: caro Class: economico Class: medio
## Sensitivity      0.800000      1.0000      1.0000
## Specificity      1.000000      0.9955      1.0000
## Pos Pred Value   1.000000      0.9980      1.0000
## Neg Pred Value   0.998622      1.0000      1.0000
## Prevalence       0.006854      0.6923      0.3009
## Detection Rate   0.005483      0.6923      0.3009
## Detection Prevalence 0.005483      0.6936      0.3009
## Balanced Accuracy 0.900000      0.9978      1.0000

```

Primero que nada, se observa que el modelo pudo predecir como se puede observar un porcentaje de 99.68%. Las predicciones para las clases economico y medio son casi perfectas, mientras que la clase caro muestra una leve disminución en la sensibilidad (80%), pero sigue siendo un buen resultado. La alta precisión y Kappa indican que el modelo está sobreajustado.

11. Compare la eficiencia del modelo anterior con los de clasificación de las entregas anteriores ¿Cuál se demoró más en procesar? ¿Cuál se equivocó más? ¿Cuál se equivocó menos? ¿por qué?

```
## [1] "Training time: 20.5764999389648"
```

De la entrega 2 del proyecto 2 se observa en terminos de tiempo de ejecución que este es el algoritmo más tardado a comparación de los modelos cruzados y no cruzados en las cuales toman un rango de tiempo de 2 a 10 segundos. En cuanto a los modelos que más se equivocaron se observa que el modelo knn tiene un porcentaje de precisión de 74.66%. Aunque se ve que hay otros modelos que estan cerca del rango de 90% a 100% como Bayes se observa que este es el mejor modelo.